Cardiff Metropolitan University | Prifysgol Metropolitan Caerdydd

School of Technologies

Assessment Brief

Module Code	Module Title
CIS6006	Cybersecurity and Cryptography
Academic Year	**Semester**
2024	2

Module Leader email

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Assessment Details

Assessment title	Abr.	Weighting
Identifying Legal, social-technical, professional and ethical dimension in cyber security and BCP / DRP for SkyMatrix Aviation Systems	PRES1	25%

Pass marks are 40% for undergraduate work and 50% for postgraduate work unless stated otherwise.

Task/assessment brief:

Scenario

SkyMatrix Aviation Systems Ltd. is a leading high-tech aviation company that develops and maintains comprehensive aviation ecosystem solutions including critical airport infrastructure systems, air traffic management software, aircraft maintenance tracking systems, and customer-facing digital services. The company operates a cloud-based platform (hosted on Microsoft Azure) that provides integrated services including online flight booking, real-time baggage tracking, personalized travel offers, loyalty program management, and mobile check-in solutions. They handle sensitive passenger data, payment information, flight operations data, aircraft maintenance records, and proprietary aviation algorithms. The company operates from three locations including a main development center and two operational hubs, with 200 employees, and provides systems to over 35 international airports, 12 major airlines, and serves approximately 2.5 million passengers annually through their digital platform.

Tasks

Now, as a senior Cyber Security analyst, your responsibility is to train and prepare a presentation to educate these newly trained cyber security officers at Sky Metrix on legal, socio-technical, professional and ethical dimensions of their duty and what they need to adhere to. LO 05)

Word count (or equivalent): **Word count – NA / 15 Miniute Presentation.**

This a reflection of the effort required for the assessment. Word counts will normally include any text, tables, calculations, figures, subtitles and citations. Reference lists and contents of appendices are excluded from the word count. Contents of appendices are not usually considered when determining your final assessment grade.

Academic or technical terms explained:

Understanding -
Student will understand how penetration testers need to work on the legal, socio-technical, professional, and ethical aspects of their duties. We'll cover laws and regulations governing cybersecurity, emphasizing the importance of compliance and proper authorization. Additionally, we'll discuss understanding human behavior, effective communication with stakeholders, ethical responsibilities, and the balance between security testing and patient welfare.

Guidelines for the Presentation format

- Submit your power point presentation file (.ppt / .pptx) to CMU moodle and after attaching your Turnitin report, submit the power point presentation file to SIS.
- Each team made up of 4 people (according to bach, better if no more than 4)
- Presentation duration : 15 minutes
- Individual Self-reflection: 3 min for each member
- Every group is expected to hand over their slides of their presentation on the presentation day
- The Group leader is expected to attach a member's contribution declaration as the cover page into this report.
- Based on the presentation and the Q&A session the panel will grade the proceedings.
- Students absent on the presentation day will not be awarded any mark irrespective of their contribution to the group work.
- Common marks given for the group presentation. <u>However, the members can get individual marks for their self-reflection activity.</u>
- Marks are allocated based on content(70%), Presentation(20%) and individual selfreflection(10%)

  Referencing should be done strictly using Harvard system
- All citation should be given as per "***Harvard Referencing Style***" for Text, Tables and Figures
- Late submission will be penalized
- Please refer to the ICBT student handbook for further details
- **Improper report structure will lead to a reduction in the total marks.**

Note: Any kind of copying and plagiarized text will lead to automatic Re-submission.

Submission Details

Submission Deadline:	This will be provided on the Moodle submission point.	**Estimated Feedback Return Date**	This will normally be 20 working days after initial submission.
Submission Time:	By 2.00pm on the deadline day.		

Moodle/Turnitin:

Any assessments submitted after the deadline will not be marked and will be recorded as a non-attempt unless you have had an extension request agreed or have approved mitigating circumstances. See the School Moodle pages for more information on extensions and mitigating circumstances.

File Format:

The assessment must be submitted as a pdf document (save the document as a pdf in your software) and submit through the Turnitin submission point in Moodle.

Your assessment should be titled with your:

student ID number, module code and assessment ID,
e.g. st12345678 BHL5007 WRIT1

Feedback

Feedback for the assessment will be provided electronically via Moodle. Feedback will be provided with comments on your strengths and the areas which you can improve. View the guidance on how to access your feedback.

All marks are provisional and are subject to quality assurance processes and confirmation at the programme Examination Board.

Assessment Criteria

Learning outcomes assessed

- LO 05 : Contextualize the wider legal, socio-technical, professional and ethical dimensions of cyber security.

Other skills/attributes developed

This includes elements of the Cardiff Met EDGE (Ethical, Digital, Global and Entrepreneurial skills) and other attributes developed in students through the completion of the module and assessment. These will also be highlighted in the module guidance, which should be read by all students completing the module. Assessments are not just a way of auditing student knowledge. They are a process which provides additional learning and development through the preparation for and completion of the assessment.

- **Ethical:**
- By conforming to the legal, socio-technical, ethical professions in penetration testing, students can hone their sense of ethics through this task. This is because students comprehend how cybersecurity law affects them and why they should avoid hacking into systems without authority so as to be compliant. Such things as understanding various behaviors of people towards others; ways communication occurs between stakeholders plus making integrity-based choices have been discussed in order to improve upon these areas.

Marking/Assessment Criteria

Contents – 70%

Task	Poor < 40%	Average 40 %- 49%	Satisfactory 50% -59%	Good 60%-69%	Excellent >70%
TASK	very limited, with many errors and gaps; little or no evidence discussed; or irrelevant and inaccurate. Student Presentation is clumsy, not correctly reflect the knowledge, difficult to follow, dull. Time Management: significantly under or over time; has clearly not tried out	Basic understanding on the theory and concepts of legal, socio-technical, professional and ethical dimensions of cyber security. Not enough discussions on the key and relevant security aspects regarding the context.	In the legal dimension, it shows an adequate grasp of cybersecurity-related legal requirements, covering most essential regulations, compliance obligations, and potential legal consequences, along with basic documentation needs. The socio-technical dimension reflects a reasonable understanding of human factors, including social engineering threats and organizational security culture, while also acknowledging the balance between usability and security. The professional dimension includes coverage of major industry standards, professional development needs, relevant certifications, and the importance of continuous learning and core competencies.	A Good level performance demonstrates comprehensive understanding across all cybersecurity dimensions with detailed analysis and practical applications. The work shows good grasp of cybersecurity legal requirements including thorough discussion of compliance obligations and comprehensive coverage of key regulations with detailed explanations. The socio-technical aspects are well-developed, featuring insightful analysis of human factors, comprehensive discussion of social engineering threats, and well-developed treatment of organizational security culture with strong connections between technical and social aspects and thoughtful consideration of usability versus security balance.	Critically analyze the legal, socio-technical, professional and ethical dimensions of your suggested system with relevant examples with references to cyber laws and international standards. Critically analyze the view of different stakeholders. Coverage of the limits and vulnerabilities of the technologies, applications, or systems being discussed.

Presentation – 20%

Task	Poor < 40%	Average 40 %- 49%	Satisfactory 50% -59%	Good 60%-69%	Excellent >70%
TASK	Students Presentation skill is clumsy, disjointed, difficult to follow, dull. Time Management: significantly under or over time; has clearly not tried out	Presentation Skills are not always clear or easy to follow; unimaginative and unengaging. significantly over time; material fairly disorganised and rushed in Time Management. Shown No any hand gustures and visual aids	Presentation Skills: conveys meaning, but sometimes unclear or clumsy; Time Management: more or less right length, but some material not covered properly as a result, OR, significantly over-runs; Group Skills: responds Low level of poor hand gustures and visual aids	Presentation Skills of the theam is generally clear, lively; use of appropriate visual aids and also in Time Management, students are well organised. Group Skills - attempts to engage with group and responds reasonably well to questions.	Student's Presentation Skills are clear, lively, imaginative; good use of visual aids. Time Management: perfectly well organised. Well planed presentation delivery.

selfreflection(10%)

Task	Poor < 40%	Average 40 %- 49%	Satisfactory 50% -59%	Good 60%-69%	Excellent >70%
TASK	Student not reflect much knowledge within the presentation. No clear idea. No any support for the group's content delivery.	Uncomfortable responding to questions; no attempt at engaging with group in delivery.	Reasonably well to questions section, but makes no real attempt to engage with group or promote discussion.	Well confidence content delivery shown by the student. Lively engage with questioning session. Properly handle the groop work and support to complete group activity providing meaningfull contents.	Student's Group Skills engages well with group members encourages discussion and responds well to questions. Show very high confidence content delivery. Show high contribution to the group presentation.

further Information

Staff note – Please check that these are aligned with your programme's expectations.

Who can answer questions about my assessment?

Questions about the assessment should be directed to the staff member who has set the task/assessment brief. This will usually be the Module Leader. They will be happy to answer any queries you have.

Staff members can often provide feedback on an assignment plan but cannot review any drafts of your work prior to submission. The only exception to this rule is for Dissertation Supervisors to provide feedback on a draft of your dissertation.

Referencing and independent learning

Please ensure you reference a range of credible sources, with due attention to the academic literature in the area. The time spent on research and reading from good quality sources will be reflected in the quality of your submitted work.

Remember that what you get out of university depends on what you put in. Your teaching sessions typically represent between 10% and 30% of the time you are expected to study for your degree. A 20-credit module represents 200 hours of study time. The rest of your time should be taken up by self-directed study.

Unless stated otherwise you must use the **HARVARD** referencing system. Further guidance on referencing can be found in the Study Smart area on Moodle and at www.citethemrightonline.com (use your university login details to access the site). Correct referencing is an easy way to improve your marks and essential in achieving higher grades on most assessments.

Technical submission problems

It is strongly advised that you submit your work at least 24 hours before the deadline to allow time to resolve any last minute problems you might have. If you are having issues with IT or Turnitin you should contact the IT Helpdesk on (+44) 2920 417000. You may require evidence of the Helpdesk call if you are trying to demonstrate that a fault with Moodle or Turnitin was the cause of a late submission.

Extensions and mitigating circumstances

Short extensions on assessment deadlines can be requested in specific circumstances. If you are encountering particular hardship which has been affecting your studies, then you may be able to apply for mitigating circumstances. This can give the teachers on your programme more scope to adapt the assessment requirements to support your needs. Extensions and mitigating circumstances policies and procedures are regularly updated. You should refer to your degree programme or school Moodle pages for information on extensions and mitigating circumstances.

Unfair academic practice

Cardiff Met takes issues of unfair practice **extremely seriously.** The University has procedures and penalties for dealing with unfair academic practice. These are explained in full in the University's Unfair Practice regulations and procedures under Volume 1, Section 8 of the Academic Handbook. The Module Leader reserves the right to interview students regarding any aspect of their work submitted for assessment.

Types of Unfair Practice, include:

Plagiarism, which can be defined as using without acknowledgement another person's words or ideas and submitting them for assessment as though it were one's own work, for instance by copying, translating from one language to another or unacknowledged paraphrasing. Further examples include:

- Use of any quotation(s) from the published or unpublished work of other persons, whether published in textbooks, articles, the Web, or in any other format, where quotations have not been clearly identified as such by being placed in quotation marks and acknowledged.
- Use of another person's words or ideas that have been slightly changed or paraphrased to make it look different from the original.
- Summarising another person's ideas, judgments, diagrams, figures, or computer programmes without reference to that person in the text and the source in a bibliography/reference list.
- Use of assessment writing services, essay banks and/or any other similar agencies (NB. Students are commonly being blackmailed after using essay mills).
- Use of unacknowledged material downloaded from the Internet.
- Re-use of one's own material except as authorised by your degree programme.

Collusion, which can be defined as when work that that has been undertaken with others is submitted and passed off as solely the work of one person. Modules will clearly identify where joint preparation and joint submission are permitted, in all other cases they are not.

Fabrication of data, making false claims to have carried out experiments, observations, interviews or other forms of data collection and analysis, or acting dishonestly in any other way.

How is my work graded?

Assessment grading is subject to thorough quality control processes. You can view a summary of these processes on the Assessment Explained Infographic.

Grading of work at each level of Cardiff Met degree courses is benchmarked against a set of general requirements set out in Volume 1, Section 4.3 of our Academic Handbook. A simplified version of these Grade Band Descriptors (GBDs) with short videos explaining some of the academic terminology used can be accessed via the Facilitation of Learning resource page.

We would strongly recommend looking at the Study Smart area of Moodle to find out more about assessments and key academic skills which can have a significant impact on your grades. Always check your work thoroughly before submission.

Module Number: CIS6006	Module Name: Cyber Security and Cryptography	Module Leader: Aruna Indika		Year/Term: 2025/2

MODULE ASSESSMENT STRATEGY		BRIEFING INCLUDES:	X	IV's COMMENTS / RECOMMENDATIONS	MODULE LEADER'S ACTION TAKEN
Presentation	*25%*	Description of Task/Problem/Topic			
Report	*75%*	Hand in Details			
		Guidance Notes			
		Learning Outcomes			
Total	***100%***	Assessment Criteria			
ASSESSMENT DETAILS		Mark/Grade Criteria			
Analyse cyber security in developing Online Shopping Mall Management System		Marking Scheme			
		Feedback Sheet			
Assessment type: WRIT1 -- Assignment		Referencing Requirements Given			

Nominated IV: Aruna Kelum	IV Signature:	Date:

CARDIFF METROPOLITAN LINK TUTOR COMMENTS:	EXTERNAL EXAMINER COMMENTS:

	Please check appropriate box below	X	Please check the appropriate box below
	I confirm that I have considered the above draft assignment/exam and I am happy to approve the content. **Assessment can now be forwarded to the External Examiner for approval.**		I confirm that I have considered the above draft assignment/exam and I am happy to approve the content. **Assessment can now be released to the students.**
	I confirm that I have considered the above draft assignment/exam and I am happy to approve the content subject to the above amendments. **Assessment can be forwarded to the External Examiner once these changes have been implemented and verified.**		I confirm that I have considered the above draft assignment/exam and I am happy to approve the content subject to the above amendments. **Assessment can be released to students once these changes have been implemented and verified.**
	I confirm that I have considered the above draft assignment/exam and suggest the above amendments. **I would like to see the final amended version before I confirm approval.**		I confirm that I have considered the above draft assignment/exam and suggest the above amendments.

				I would like to see the final amended version before I confirm approval.	
Cardiff Metropolitan Link Tutor:	Link tutor signature:	Date:	External Examiner:	EE Signature:	Date:

Module Title	Module Number	JACS Subject Code(s) and % of each subject	ASC Category(ies)
Cyber Security and Cryptography	CIS6006	I100	6

Level (3 to 8)	Credits	*ECTS Credit*	Module Value *(1=20 credits)*	% Taught in Welsh	Module Type
6	20	10	1.0	0%	Taught

Teaching Period *(Term/Semester)*	Pre-requisites
Term 2	None

Module Leader	School(s)	Campus
Abdelrahman Abuarqoub	Cardiff School of Technologies	Llandaff

Assessment Methods

Assessment Code and Method	Duration/Length of Assessment Method	Weighting of Assessment	Threshold	Approximate Date of Submission
PRES1 -- Group Presentation	15 minutes (1,000 word equivalent)	25%	1	Mid Module
WRIT1 -- Assignment	3,000 word equivalent	75%	1	End of Module

Aim(s)

This module aims to encourage students to critically reflect on a range of information security concepts and topical issues including information security, cryptography, cryptoanalysis and secure systems. Students will have the opportunity to critically analyse and design practical applications using modern cryptography.

Learning Outcomes

On successful completion of the module, students should be able to:

- Identify and evaluate information security risks in a variety of real world environments.
- Critically evaluate essential components and principles of cryptography
- Demonstrate an understanding of and justify the use of appropriate security and cryptographic techniques for the design and implementation of secure systems.
- Design and implement symmetric and public-key encryption for real-world applications
- Contextualise the wider legal, socio-technical, professional and ethical dimensions of cyber security.

Learning and Teaching Delivery Methods

Method	Rationale	Type of Contact (scheduled/ guided independent study/placement)	Total hours
Studios	To enable core knowledge and understanding content to be delivered to the whole module cohort	Scheduled	24

Workshops	To allow exploration of all aspects of module content (knowledge, understanding, skills & other attributes) in an interactive group setting	Scheduled	24
Independent Study	To enable students to independently develop their understanding of the module concepts and to complete formative & summative assessment activity	Guided Independent Study	152
			200

Indicative Content
Identification and authentication, confidentially, integrity and availability, access control techniques. Risk management. Security models and evaluation. Physical and environmental security. An introduction to cryptography: Principles of cryptography; Confidentiality, integrity, non-repudiation, data origin authentication and entity authentication; A history of cryptography; Basic information theory; Essential number theory, algebra and discrete mathematics; Classical, symmetric and asymmetric encryption techniques and algorithms; Steganography; Cryptanalysis; Encryption and decryption, types of cipher, symmetric and asymmetric key systems, message digests and signatures. The application of cryptography: An overview of the application of cryptography; Encrypting data at rest; Digital signatures; Digital certificates; Certification authorities; Public-Key Infrastructure (PKI); Securing email and Internet communications (SSL/TLS); Virtual Private Networks (VPNs); The provision for data origin authentication, entity authentication and non-repudiation in network communications; Cryptanalysis and cryptographic backdoors; The future of cryptography. Cryptaanalysis, steganaography, quantum cryptography

Required Reading
Keith, M (2017) Everyday Cryptography: Fundamental Principles and Applications, OUP Oxford; 2nd edition.

Recommended Reading
Vacca, J (2013) Cyber Security and IT Infrastructure Protection, Syngress Mooney, T, (2015) Information Security: A Practical Guide, IT Governance Publishing Alexander, D., Finch, A., Sutton, D. and Taylor, A. (2020) Information Security Management Principles (3rd ed.), BCS Bruce, S (2015) Applied Cryptography: Protocols, Algorithms and Source Code in C, – John Wiley & Sons; 20th Anniversary edition.

Access to Specialist Requirements
Demonstration capability for modern cryptographic algorithms